

A Point-Centered Effective Model of Inverse-Square Intensity-Gradient Transport

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Abstract

We study an effective exterior field

$$\Psi(\mathbf{x}, t) = Ar^{-p}f(\theta, \phi)e^{i\Phi(r, \theta, \phi, t)}, \quad r \geq r_{\min} > 0,$$

and interpret $I = |\Psi|^2$ operationally as an intensity field. For an angular modulation f that is independent of r , the angle-averaged radial gradient obeys $\langle |\partial_r I| \rangle_\Omega \propto r^{-(2p+1)}$, so the choice $p = \frac{1}{2}$ produces inverse-square scaling. We emphasize that this law is selected by the assumed envelope and is not derived from a deeper wave equation. A probe coupled through $U_{\text{probe}} = -\mu I$ experiences force $\mathbf{F} = \mu \nabla I$. In the overdamped regime $\Gamma \dot{\mathbf{x}} = \mu \nabla I$, radial motion is strictly inward because $\dot{r} = (\mu/\Gamma)\partial_r I < 0$. Inertial motion with linear damping is governed by $M\ddot{\mathbf{x}} + \gamma\dot{\mathbf{x}} = \mu \nabla I$ and follows from a Lagrangian with Rayleigh dissipation. Numerical integrations of eight probes launched from $r_0 = 4.8$ show inward convergence in both overdamped and damped inertial regimes over a finite time horizon. The result is a mathematically explicit effective model of point-centered inverse-square intensity-gradient transport, not a theory of gravity.

1 Introduction

Earlier work on radially weighted spiral interference suggested a route to inverse-square inward transport in two spatial dimensions. A naive three-dimensional cylindrical extension, however, remains axis-centered: it attracts probes toward a line rather than toward a point. The central geometric question is therefore how to build a field whose transport is point-centered from the outset.

The purpose of the present paper is to formulate and test a minimal spherical model. Rather than extruding a planar pattern into three dimensions, we posit an exterior envelope whose amplitude depends on the full spherical radius and whose angular structure is defined on the sphere. This gives inward gradients in every direction and makes the radial scaling transparent.

The scope is deliberately narrow. We do not derive gravity, mass density, spacetime curvature, or Einstein dynamics. We also do not derive the envelope from a deeper field equation. Instead, we study the transport consequences of a specific effective exterior field and ask whether it yields a clean and numerically robust mechanism for point-centered inverse-square inward motion.

2 Effective Spherical Field Model

We work in spherical coordinates (r, θ, ϕ) with

$$r = \sqrt{x^2 + y^2 + z^2}. \tag{1}$$

We define the complex field

$$\Psi(r, \theta, \phi, t) = Ar^{-p} f(\theta, \phi) e^{i\Phi(r, \theta, \phi, t)}, \quad (2)$$

where $A > 0$ sets the amplitude scale, $p > 0$ controls radial attenuation, $f(\theta, \phi)$ is a bounded angular modulation, and Φ is a phase function.

For the numerical realization used here, we choose

$$f(\theta, \phi) = 1 + 0.18 \sin \theta \cos \phi + 0.12 \sin^2 \theta \cos(2\phi), \quad (3)$$

which is strictly positive and introduces mild anisotropy without changing the radial structure. The intensity is

$$I = |\Psi|^2 = A^2 r^{-2p} |f(\theta, \phi)|^2. \quad (4)$$

In this paper we focus on the special case

$$p = \frac{1}{2}, \quad (5)$$

for which

$$I \propto r^{-1}, \quad \partial_r I \propto -r^{-2}. \quad (6)$$

Two clarifications are important. First, the phase $\Phi(r, \theta, \phi, t)$ does not enter the present transport law. It is retained only to indicate how the envelope might later be embedded in a more complete wave-dynamical theory. The present model is intensity-driven rather than phase-driven.

Second, the choice $p = \frac{1}{2}$ is not claimed to be the asymptotic free-space decay law of an ordinary three-dimensional radiating solution. It is adopted here as an effective exterior envelope because it yields an inverse-square intensity gradient. The current paper therefore studies the consequences of this envelope once posited, rather than deriving it from microscopic wave physics.

3 Analytic Radial Scaling Law

The spherical gradient of the intensity is

$$\nabla I = \hat{\mathbf{r}} \partial_r I + \hat{\boldsymbol{\theta}} \frac{1}{r} \partial_\theta I + \hat{\boldsymbol{\phi}} \frac{1}{r \sin \theta} \partial_\phi I. \quad (7)$$

Differentiating (4) with respect to r gives

$$\partial_r I = -2p A^2 r^{-(2p+1)} |f(\theta, \phi)|^2. \quad (8)$$

Since $|f(\theta, \phi)|^2 > 0$, the radial component is inward everywhere on the exterior domain.

To separate the power-law scaling from the angular anisotropy, define the spherical average

$$\langle g \rangle_\Omega = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi g(\theta, \phi) \sin \theta \, d\theta \, d\phi. \quad (9)$$

Applying this to (8) yields

$$\langle |\partial_r I| \rangle_\Omega = 2p A^2 r^{-(2p+1)} \langle |f(\theta, \phi)|^2 \rangle_\Omega. \quad (10)$$

Thus the angular modulation changes only the prefactor, not the exponent:

$$\langle |\partial_r I| \rangle_\Omega \propto r^{-(2p+1)}. \quad (11)$$

For $p = \frac{1}{2}$,

$$\langle |\partial_r I| \rangle_\Omega \propto r^{-2}. \quad (12)$$

Because the envelope scales as $r^{-1/2}$, both the intensity and its gradient diverge as $r \rightarrow 0$. The present model should therefore be interpreted as an effective exterior field defined only on

$$r \geq r_{\min} > 0. \quad (13)$$

Any physical realization would require regularization of the core.

4 Probe Coupling and Dynamical Laws

We couple a probe to the field through the interaction energy

$$U_{\text{probe}}(\mathbf{x}) = -\mu I(\mathbf{x}), \quad \mu > 0. \quad (14)$$

The resulting force is

$$\mathbf{F}_{\text{probe}} = -\nabla U_{\text{probe}} = \mu \nabla I. \quad (15)$$

Because $\partial_r I < 0$, the radial component of $\mathbf{F}_{\text{probe}}$ points inward.

The inertial dynamics are

$$M \ddot{\mathbf{x}} = \mu \nabla I, \quad (16)$$

or, with linear damping,

$$M \ddot{\mathbf{x}} + \gamma \dot{\mathbf{x}} = \mu \nabla I. \quad (17)$$

In the overdamped limit we write

$$\Gamma \dot{\mathbf{x}} = \mu \nabla I, \quad (18)$$

where $\Gamma > 0$ is a mobility denominator (drag coefficient).

Proposition 1. *Assume $p > 0$ and that the angular modulation $f(\theta, \phi)$ is independent of r . Then every solution of the overdamped law (18) is strictly inward while it remains in the exterior domain: if $r(t) > r_{\min}$, then $\dot{r}(t) < 0$.*

Proof. Take the radial component of (18):

$$\dot{r} = \hat{\mathbf{r}} \cdot \dot{\mathbf{x}} = \frac{\mu}{\Gamma} \hat{\mathbf{r}} \cdot \nabla I = \frac{\mu}{\Gamma} \partial_r I. \quad (19)$$

Using (8),

$$\dot{r} = -\frac{2p\mu A^2}{\Gamma} r^{-(2p+1)} |f(\theta, \phi)|^2 < 0, \quad (20)$$

for every $r > r_{\min}$. Hence the radius decreases monotonically until the trajectory reaches the cutoff or the integration horizon. $\square$

This analytic result is stronger than any finite set of overdamped trajectory plots: inward motion in the overdamped regime is a theorem of the model, while the overdamped numerics are best read as visualization and implementation checks.

5 Dissipative Formulation

The natural overdamped variational statement is dissipative rather than conservative. Let

$$V(\mathbf{x}) = -\mu I(\mathbf{x}). \quad (21)$$

Then the overdamped law (18) follows from minimizing the Rayleighian

$$\mathcal{R}(\dot{\mathbf{x}}; \mathbf{x}) = \frac{\Gamma}{2} |\dot{\mathbf{x}}|^2 + \nabla V(\mathbf{x}) \cdot \dot{\mathbf{x}}. \quad (22)$$

Variation with respect to $\dot{\mathbf{x}}$ gives

$$\frac{\partial \mathcal{R}}{\partial \dot{\mathbf{x}}} = \Gamma \dot{\mathbf{x}} + \nabla V(\mathbf{x}) = 0, \quad (23)$$

and therefore

$$\Gamma \dot{\mathbf{x}} = -\nabla V(\mathbf{x}) = \mu \nabla I, \quad (24)$$

which is precisely (18). In this formulation the free energy decreases monotonically along trajectories:

$$\dot{V} = \nabla V \cdot \dot{\mathbf{x}} = -\Gamma |\dot{\mathbf{x}}|^2 \leq 0. \quad (25)$$

Equivalently,

$$\frac{d}{dt} I(\mathbf{x}(t)) = \frac{\mu}{\Gamma} |\nabla I|^2 \geq 0, \quad (26)$$

so the probe climbs the intensity landscape while dissipating free energy.

For inertial motion, introduce the effective Lagrangian

$$L = \frac{M}{2} |\dot{\mathbf{x}}|^2 - V(\mathbf{x}) = \frac{M}{2} |\dot{\mathbf{x}}|^2 + \mu I(\mathbf{x}), \quad (27)$$

and the Rayleigh dissipation function

$$\mathcal{D} = \frac{\gamma}{2} |\dot{\mathbf{x}}|^2. \quad (28)$$

The Euler-Lagrange equation with dissipation yields

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{x}}} - \frac{\partial L}{\partial \mathbf{x}} + \frac{\partial \mathcal{D}}{\partial \dot{\mathbf{x}}} = 0, \quad (29)$$

that is,

$$M \ddot{\mathbf{x}} + \gamma \dot{\mathbf{x}} = \mu \nabla I, \quad (30)$$

which matches (17). Thus both overdamped and damped inertial motion admit consistent dissipative formulations.

6 Numerical Methods

The scaling test evaluates (10) on the domain $r \in [0.4, 6.0]$ using 500 radial samples. At each radius, the spherical average is estimated from 2000 directions sampled uniformly on the sphere. The numerical scaling plot is therefore an implementation check of the analytic law, not an independent discovery.

For trajectory calculations, eight probes are initialized on a sphere of radius

$$r_0 = 4.8 \quad (31)$$

at distinct angular positions. The trajectories shown below were integrated to total time

$$T = 70 \tag{32}$$

with timestep

$$dt = 0.005. \tag{33}$$

The singular core is excluded by imposing the regularization floor

$$r_{\min} = 0.12. \tag{34}$$

The overdamped system is evolved with a first-order explicit update using $\Gamma = 1$ and $\mu = 1$. For the chosen timestep and finite horizon, this scheme is stable and sufficient for the first-order flow. Because inward overdamped motion already follows analytically from the proposition above, these simulations are included mainly as visualizations and consistency checks.

The inertial system (17) is integrated with fourth-order Runge-Kutta using

$$M = 1, \quad \mu = 1, \quad \gamma = 2. \tag{35}$$

The specific trajectories and final radii depend on the finite integration horizon, the damping constant, the mobility coefficient, and the core cutoff $r_{\min}$. The robust numerical claim is therefore qualitative inward convergence rather than any universal terminal radius.

7 Results

7.1 Scaling Verification

The analytic law predicts

$$\beta = 2p + 1, \tag{36}$$

so for $p = \frac{1}{2}$ one expects $\beta = 2$. Fitting the mean radial gradient magnitude to

$$\langle |\partial_r I| \rangle_\Omega \sim r^{-\beta} \tag{37}$$

gives

$$\beta = 2.00000, \quad R^2 = 1.00000000, \tag{38}$$

with relative error at the level of machine precision. The scaling computation therefore confirms correct implementation of the assumed envelope.

7.2 Intensity Structure

Figure 2 shows an equatorial slice of the intensity field. The dominant structure is point-centered, with the $1/r$ envelope clearly visible and the angular modulation appearing only as mild anisotropy.

7.3 Inward Trajectories

Figure 3 summarizes the probe trajectories. Over the finite time horizon $T = 70$, all eight overdamped trajectories move inward, consistent with the monotonicity result above, with mean final radius

$$\bar{r}_f = 0.267. \tag{39}$$

All eight inertial trajectories with damping also move inward, with mean final radius

$$\bar{r}_f = 1.383. \tag{40}$$

In both cases the inward fraction is $8/8$.

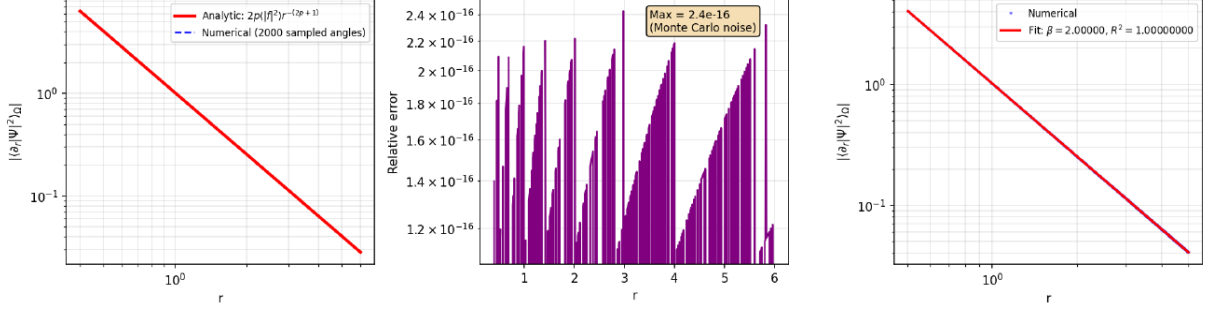


Figure 1: Radial-scaling verification for the spherical model. Panel (a) compares the analytic and numerical mean radial gradients, panel (b) shows the numerical-analytic relative error, and panel (c) shows the power-law fit confirming $\beta = 2p + 1$. For $p = \frac{1}{2}$, the fit gives $\beta = 2.00000$ with $R^2 = 1.00000000$, confirming inverse-square scaling to numerical precision.

Table 1: Finite-time trajectory outcomes for the spherical model. The reported mean final radii depend on the chosen time horizon, damping, mobility, and core cutoff; the robust claim is that all tested trajectories move inward.

Regime	Mean initial radius	Mean final radius	Inward fraction
Overdamped	4.800	0.267	8/8
Inertial with damping	4.800	1.383	8/8

7.4 Role of Regularization and Parameter Dependence

The strongest inward contraction occurs near the core, where the exterior envelope is singular. For that reason, the reported finite-time radii should not be interpreted as universal asymptotic values. They depend on the chosen cutoff $r_{\min}$, damping γ , mobility or coupling strength, and total integration time.

What is robust in the present computations is the sign of the motion: across the tested initial conditions and in both overdamped and inertial regimes, all probes move inward. The numerics therefore support a qualitative transport claim - persistent inward convergence - rather than a cutoff-independent statement about late-time collapse.

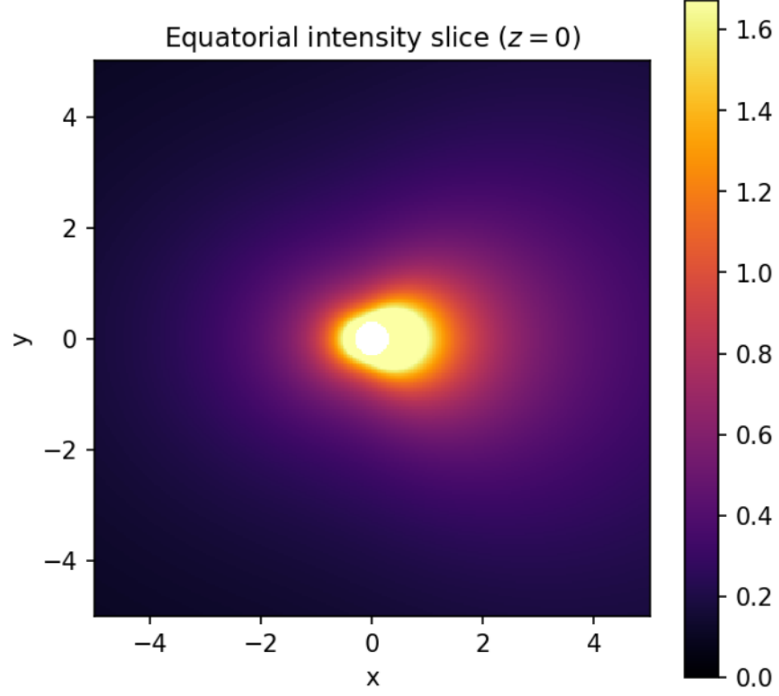


Figure 2: Equatorial intensity slice ($z = 0$) of the spherical model. The dominant radial envelope is point-centered, while the angular modulation adds mild directional structure without changing the inward radial character of the field.

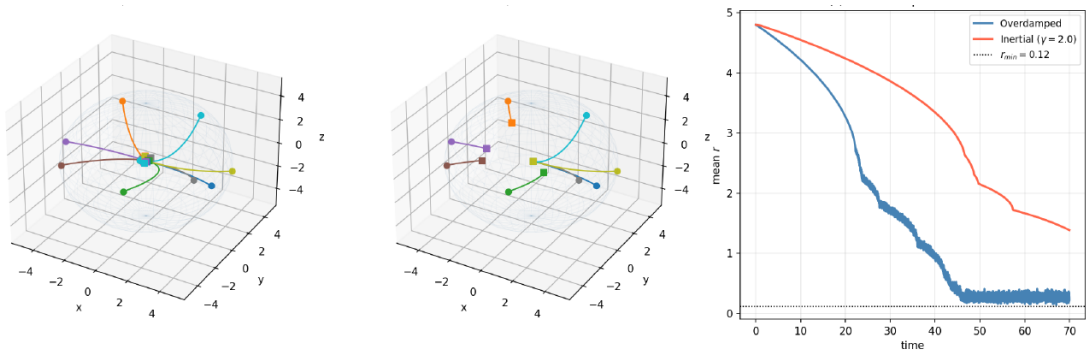


Figure 3: Trajectory dynamics in the spherical model. Panel (a) shows overdamped motion, panel (b) shows inertial motion with damping ($\gamma = 2$), and panel (c) shows the mean radial position versus time. Both regimes exhibit inward transport over the full integration interval, although the rate of contraction depends on the dynamical law and parameter choices.

8 Scope and Claims

The paper establishes the following.

1. For the field ansatz (2), the angle-averaged radial intensity gradient scales analytically as $r^{-(2p+1)}$ whenever $f(\theta, \phi)$ is independent of r .
2. For $p = \frac{1}{2}$, this gives an inverse-square radial law.
3. The probe coupling $U_{\text{probe}} = -\mu I$ produces inward motion in all directions for a point-centered field.
4. In the overdamped regime, inward radial motion is an analytic consequence of the model, not merely a numerical observation.
5. The damped inertial system also exhibits inward convergence from all tested initial conditions over the reported finite time horizon.

At the same time, the model does not derive the envelope from a deeper field equation, does not establish Newtonian gravity from mass density, and does not produce spacetime curvature or Einstein dynamics. It is best understood as a minimal effective model of point-centered inverse-square intensity-gradient transport.

9 Toward a Testable Physical System

A possible experimental analog would require

1. a localized wave source or engineered field producing an approximately spherical exterior intensity envelope,
2. controlled angular modulation of that envelope, and
3. probes whose motion depends on local intensity gradients.

The clearest first platform is an optical intensity landscape. Optical systems allow deliberate shaping of radial envelopes, direct control of angular anisotropy, and quantitative measurement of intensity-gradient forces on probe particles. In such a setup, a point-centered optical envelope with controlled angular modulation would provide the most direct test of the model's central prediction: inward transport toward a single intensity maximum with an approximately inverse-square radial gradient.

Other candidate platforms include acoustic fields and analog wave media with effective trapping forces. The central experimental prediction is the same: probes coupled to the field should move inward toward a point-centered region of maximum intensity with an approximately inverse-square radial gradient.

10 Discussion

The main achievement of the spherical formulation is geometric: it replaces axis-centered transport with point-centered transport. The analytic structure is also cleaner than in the cylindrical case, because the radial exponent follows immediately from the chosen envelope and the overdamped inward-motion statement can be proved directly.

The main weakness of the model is equally clear. The inverse-square behavior is built into the effective envelope rather than emerging from a deeper dynamical theory. The present paper therefore does not solve the emergence problem; it isolates an explicit transport mechanism and studies its consequences. This is useful, but more modest than a first-principles derivation.

A second limitation is the near-core singularity. The strong late-time contraction is influenced by the exterior-field divergence and by the imposed cutoff. A more physical successor model would regularize the core explicitly or derive a finite inner structure from the field equation itself.

A third limitation is that phase is passive here. Since the transport law depends only on $|\Psi|^2$, the present framework is better described as an intensity-gradient model than as a genuinely phase-dynamical coherence theory. Extending the model to a field whose phase evolves and back-reacts on the transport remains an open next step.

11 Conclusion

We have formulated an effective spherical model of point-centered inward transport based on the field

$$\Psi(\mathbf{x}, t) = Ar^{-p} f(\theta, \phi) e^{i\Phi(r, \theta, \phi, t)}. \quad (41)$$

For r -independent angular modulation, the angle-averaged radial intensity gradient scales analytically as $r^{-(2p+1)}$, so the case $p = \frac{1}{2}$ gives an inverse-square law. A probe coupled through $U_{\text{probe}} = -\mu I$ moves inward under both overdamped and inertial dynamics, and the overdamped radial monotonicity can be proved directly from the model.

The main result is therefore precise but limited: the paper provides a mathematically explicit, numerically illustrated, point-centered effective model of inverse-square intensity-gradient transport. It is not a theory of gravity, but it is a controlled step toward understanding how gravity-like radial transport can be mimicked by structured fields.

References

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